



Research paper

Vertical variation in wood CO₂ efflux is not uniformly related to height: measurement across various species and sizes of Bornean tropical rainforest trees

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Limited knowledge about vertical variation in wood CO₂ efflux (R_{wood}) is still a cause of uncertainty in R_{wood} estimates at individual and ecosystem scales. Although previous studies found higher R_{wood} in the canopy, they examined several tree species of similar size. In contrast, in the present study, we measured vertical variation in R_{wood} for 18 trees including 13 species, using a canopy crane for a more precise determination of the vertical variation in R_{wood} , for various species and sizes of trees in order to examine the factors affecting vertical variation in R_{wood} and thus, to better understand the effect of taking into account the vertical and inter-individual variation on estimates of R_{wood} at the individual scale. We did not find any clear pattern of vertical variation; R_{wood} increased significantly with measurement height for only one tree, while it decreased for two more trees, and was not significantly related with measurement height in 15 other trees. Canopy to breast height R_{wood} ratio was not related to diameter at breast height or crown ratio, which supposedly are factors affecting vertical variation in R_{wood} . On average, R_{wood} estimates at individual scale, considering inter-individual variation but ignoring vertical variation, were only 6% higher than estimates considering both forms of variation. However, estimates considering vertical variation, while ignoring inter-individual variation, were 13% higher than estimates considering both forms of variation. These results suggest that individual measurements at breast height are more important for estimating R_{wood} at the individual scale, and that any error in R_{wood} estimation at this scale, due to the absence of any more measurements along tree height, is really quite negligible. This study measured various species and sizes of trees, which may be attributed to no clear vertical variation because factors causing vertical variation can differ among species and sizes.

Keywords: carbon cycling, dipterocarp, scaling, stem respiration, woody tissue respiration.

Introduction

Wood CO₂ efflux (R_{wood}), which is CO₂ originating mainly from woody tissue respiration in stems and branches of trees, is an important component of carbon cycling, observed only in forest ecosystems. Trees allocate up to 17% of fixed carbon for woody tissue respiration to sustain the growth and maintenance of the stem and branches for access to sunlight (Litton et al. 2007). Consequently, R_{wood} contributes 13–27% of ecosystem respiration (Chambers et al. 2004, Malhi et al. 2009). Nevertheless,

and despite its obvious importance, studies on the estimation of R_{wood} at the individual and ecosystem scales, as compared with other CO₂ effluxes, such as soil respiration or root and foliage respiration, are scarce (e.g., Sotta et al. 2004, Cavaleri et al. 2008, Katayama et al. 2009).

Estimation of R_{wood} at the individual and ecosystem scales requires examination of intra- and inter-individual variation in R_{wood} , because it can be highly variable at both levels. Inter-

individual variation in R_{wood} at breast height, based on surface area, has been relatively well studied and explained by stem growth rate (Ryan et al. 1994, Robertson et al. 2010), stem diameter (Meir and Grace 2002), sapwood volume (Rowland et al. 2018) and nitrogen content of woody tissue (Maier 2001, Vose and Ryan 2002). Even in a tropical rainforest, where tree size and species are diverse, the inter-individual variation was explained by those factors (Cavaleri et al. 2006, Katayama et al. 2016). However, intra-individual variation, which is the same as vertical variation in the present study, has been much less studied, because of the difficulty in accessing the canopy. The limited data available still causes uncertainty in spatial scaling.

Vertical variation in R_{wood} is more complex as compared with inter-individual variation and different results in this regard have been reported. Thus, it has been generally reported that, on a surface area basis, R_{wood} increases with height (Yoda et al. 1965, Damesin et al. 2002, Araki et al. 2010, Tarvainen et al. 2014, Han et al. 2017, Martínez-García et al. 2017); however, some studies reported that it did not vary (Ryan et al. 1996), while still others reported that it decreased with stem height (Asao et al. 2015). Tree architecture may explain such erratic behavior. For example, the ratio of annual whole-stem respiration estimated assuming constant R_{wood} along height, to annual whole-stem respiration estimated by taking into account the vertical variation in R_{wood} , was negatively related to crown ratio, i.e., the ratio of crown length to total tree height. This finding suggested that the canopy to breast height R_{wood} increased with canopy length (Araki et al. 2010). In the tropical rainforest, the ratio reportedly increased with diameter at breast height (DBH, Katayama et al. 2014). However, no consistent patterns of vertical variation in R_{wood} or of factors affecting such variation have been elucidated up to present.

Recent studies reported that vertical variation in R_{wood} caused an estimate error at the individual scale. It was found that R_{wood} increased with stem height and that only measurements at breast height underestimated annual stand-level R_{wood} by 15–20% in a Norway spruce stand (Tarvainen et al. 2014) and by 41% in a Spanish black pine forest (Martínez-García et al. 2017). These studies suggested that vertical variation in individual R_{wood} can cause considerable errors in R_{wood} estimates at the stand scale, errors that may be much higher in mature tropical rainforests, where there are many tall trees. In contrast, Katayama et al. (2014) found that the lack of R_{wood} measurements other than at breast height did not cause any bias in stem CO₂ efflux scaling for some large trees in a tropical rainforest. Nevertheless, since previous studies conducted in tropical rainforests examined vertical variation in R_{wood} for only a narrow range of tree size and species (Cavaleri et al. 2006, Katayama et al. 2014, Asao et al. 2015), it is necessary to examine vertical variation for a wider range of tree size and species in a tropical rainforest.

Therefore, to determine whether vertical measurement of R_{wood} is necessary for an accurate estimate of R_{wood} at the individual scale or not in a mature tropical rainforest, we used a canopy crane to measure the vertical variation in R_{wood} for 18 live trees from 13 tree species, with diameters of 21.3–102.6 cm and heights of 25.4–52.6 m in a Bornean tropical rainforest. Our objectives were (i) to clarify vertical variation in R_{wood} in various species of trees of different sizes, (ii) to examine whether vertical variation in R_{wood} can be explained by crown ratio or DBH and (iii) to assess how the vertical variation in R_{wood} affects the estimate of R_{wood} at the individual scale.

Materials and methods

Study site

The study was carried out in Lambir Hills National Park, Sarawak, Malaysia (4°12'N, 114°02'E) on the island of Borneo, at ~20 km southwest of Miri, Sarawak. The lowland mixed-dipterocarp forest, which is a typical mature, lowland tropical rainforest type in Southeast Asia (e.g., Ashton and Hall 1992, King et al. 2009), covers 85% of this 6949-ha park. Between 2000 and 2009, the mean annual temperature at the site was 25.8 °C and the mean annual rainfall, distributed throughout the year, was 2600 mm (Kume et al. 2011). Monthly rainfall between 1958 and 2001 averaged more than 100 mm (Kumagai et al. 2005).

An 85-m high crane with a 75-m long arm was constructed in the park for long-term ecological studies. Measurements in the present study were conducted using this mechanical device.

Measurements

Measurements of R_{wood} were conducted in 18 trees belonging to 13 species (Table 1). The trees were selected to consist of different sizes and species. Several trees of *Shorea beccariana* Burck were selected because they were large trees, which were accessible by the crane. The measurements were conducted once for each tree. R_{wood} for most trees was measured at three different heights, i.e., at breast height, below the canopy and in the canopy. In tree Nos. 11, 12 and 18, R_{wood} was measured only at breast height and in the canopy because of the limitation of accessibility. In the present study, 'the canopy' was defined as the tree area above the lowest living branch. R_{wood} was measured on 2–5 July 2014, 21–22 December 2014 and 7–9 March 2015.

As Zhao et al. (2018) found no significant effects of stem position on R_{wood} over months in *Larix principis-rupprechtii* Mayr stands measured between May and October, here we assumed that different dates of measurement did not significantly influence the vertical profile of R_{wood} in the trees under study.

It took ~2 h to measure R_{wood} for each tree and the vertical measurements were conducted at various times of day between 10:00 h and 16:00 h. Admittedly, this might affect the pattern of vertical variation in R_{wood} , as recent studies have shown that CO₂ respired by woody tissue is often transported by sap flow

Table 1. Tree species, DBH, and tree, branch and R_{wood} measurement height.

Tree no.	Tree species	DBH (cm)	Tree height (m)	The lowest living branch height (m)	Measurement height (m)		
					Breast height	Below the canopy	In the canopy
1	EBENACEAE <i>Diospyros dictyoneura</i>	21.3	30.6	25.0	1.3	22.7	25.7
2	LAURACEAE <i>Endiandra clavigera</i>	22.7	28.0	22.1	1.3	18.1	24.2
3	SAPOTACEAE <i>Palaquium</i> sp.	24.9	26.3	21.2	1.2	11.4	21.6
4	MORACEAE <i>Artocarpus anisophyllus</i>	26.1	25.4	21.3	1.2	18.0	22.5
5	IXONANTHACEAE <i>Allantospermum borneense</i>	26.5	27.7	16.5	1.2	11.7	20.3
6	CORNACEAE <i>Mastixia</i> sp.	32.0	34.3	21.2	1.2	18.7	23.0
7	DIPTEROCARPACEAE <i>Dipterocarpus globosus</i>	34.3	39.5	28.0	1.1	27.5	34.0
8	RHIZOPHORACEAE <i>Gynotroches axillaris</i>	37.8	32.0	19.2	1.1	19.0	28.0
9	DIPTEROCARPACEAE <i>Shorea beccariana</i>	49.4	43.2	34.2	1.2	33.8	37.7
10	ACHARIACEAE <i>Hydnocarpus castanea</i>	51.1	36.0	24.2	1.2	23.9	28.7
11	DIPTEROCARPACEAE <i>Shorea beccariana</i>	57.0	44.5	28.9	0.9	–	32.4, 36.4, 40.3
12	DIPTEROCARPACEAE <i>Shorea acuta</i>	57.5	47.1	25.9	1.2	–	34.8, 28.3, 34.8
13	GENTIANACEAE <i>Fagraea caudata</i>	63.7	34.9	18.0	1.2	17.9	32.3
14	ANACARDIACEAE <i>Swintonia</i> sp.	65.6	45.6	25.8	1.2	21.2	40.2
15	DIPTEROCARPACEAE <i>Shorea beccariana</i>	88.7	44.0	15.6	1.2	14.4	40.0
16	DIPTEROCARPACEAE <i>Shorea beccariana</i>	101.3	52.6	38.0	1.2	37.8	48.0
17	DIPTEROCARPACEAE <i>Shorea beccariana</i>	101.8	49.3	>34	1.2	33.9	45.4
18	DIPTEROCARPACEAE <i>Shorea beccariana</i>	102.6	47.3	30.3	1.2	–	42.7

and released at a different place from where it was produced (McGuire et al. 2007, Teskey et al. 2008); furthermore, diurnal and vertical variations in R_{wood} were found to be affected by sap flow velocity (Martin et al. 1994, McGuire and Teskey 2004). To examine the effect of diurnal variation in environmental factors as proxies of sap flow on R_{wood} , diurnal measurement of vertical variation in R_{wood} was conducted for one tree (*S. beccariana*, DBH = 69.0 cm, b1 tree of Katayama et al. 2014); R_{wood} was measured at 1.3, 10, 20, 30, 35 and 40 m height. Vertical measurements started at 9:20 on 26 June; 8:10, 10:10, 12:10, 14:00, 16:00 on 27 June; and 8:50 on 28 June in 2011 from the canopy to the ground.

R_{wood} was measured using an infrared gas analyzer (GMP343, Vaisala, Helsinki, Finland) attached inside a chamber in a closed-static system configuration. The chamber and collars were made of polyvinyl chloride. A small 12-V battery-operated fan was installed to circulate air inside the chamber. The collars were attached to stems using nondrying clay. Leakage was checked by breathing around the chamber before the measurements. An incubation period ranging from 60 to 120 s was used in each measurement. CO_2 concentration within the chamber was measured every 5 s and recorded by a monitor (M170,

Vaisala). $R_{\text{wood_measured}}$ was calculated on a surface area basis using the following equation:

$$R_{\text{wood_measured}} = \frac{V}{22.4 \times A} \left(\frac{273.2}{273.2 + q} \right) \frac{dc}{dt} \quad (1)$$

where V is the chamber volume (0.389 l), q is the air temperature inside the chamber ($^{\circ}\text{C}$), A is the projected area of the chamber ($7.854 \times 10^{-3} \text{ m}^2$) and dc is the increment of CO_2 concentration during the incubation time (dt). Equation (1) assumes that R_{wood} is measured under standard barometric pressure. All measurements at each point on the stem of each tree were replicated two or three times and averaged.

To eliminate the effects of temperature, R_{wood} was corrected to a reference temperature of 25°C using the following equation and the value of Q_{10}

$$R_{\text{wood}} = R(T)Q_{10}^{(25-t)/10} \quad (2)$$

where R_{wood} is R_{wood} at 25°C , $R(T)$ is $R_{\text{wood_measured}}$ at the stem surface temperature of T ($^{\circ}\text{C}$) and Q_{10} is a temperature coefficient of R_{wood} . Because Q_{10} was not calculated from the data in this study, it was assumed that Q_{10} was equal to 2.0, which is

the value used in a study by Cavaleri et al. (2006) in tropical rainforests in La Selva, Costa Rica.

Air temperature and stem surface temperature were measured with a thermometer (TR-52i; T&D Co. Ltd, Nagano, Japan). Stem height and diameter at R_{wood} measurement points were measured using a measuring tape. Stem height at the lowest living branch was also measured using the tape. Crown ratio was calculated as the ratio of the difference between tree height and the stem height at the lowest living branch to tree height. DBH was measured in July 2014.

Environmental observations were incorporated into the analysis of diurnal variation in R_{wood} . Air temperature was monitored on the crane at 75 m above the ground using a thermohygrometer (HMP35A, Visala). Solar radiation (W m^{-2}) was monitored using a solar radiometer (MS401, EKO, Tokyo, Japan) at the top of the crane (85.8 m above the ground). The data were logged using a programmable data logger (CR10X, Campbell Scientific, Logan, UT, USA).

Analysis of vertical variation in R_{wood}

To examine whether R_{wood} varied with height or not, first, the relationship between R_{wood} and height within an individual tree was analyzed using a linear regression model. For one tree in our list (i.e., *S. beccariana*, No. 18), this test was not conducted, because only two heights were measured. Linear regression analyses of R_{wood} with measurement height, diameter and $\ln$ (diameter) were also conducted across all the trees. To examine a factor affecting vertical variation in R_{wood} , the ratio of R_{wood} in the canopy of a tree against R_{wood} at breast height (k) was calculated for each tree and the relationship of k with crown ratio and DBH was examined. The data obtained by Katayama et al. (2014) was pooled with our own data for this analysis. The pooled data were collected from three trees of *S. beccariana* and four trees of *Dryobalanops aromatica* C.F. Gaertn., with diameters of 60–148 cm.

To examine effect of environmental factors on diurnal variation in vertical variation, three kinds of analyses were conducted. First, two-way ANOVA was conducted using the data of the diurnal measurement for the *S. beccariana* tree: R_{wood} was set as objective variable for this analysis. Height and either air temperature or solar radiation monitored at the crane top were set as explanatory variables. Height was regarded as a categorical variable. If the environmental factors affect the pattern of vertical variation in R_{wood} , then the corresponding interaction effect should be significant. Second, linear regression analysis for each height was conducted using the same data. Here, R_{wood} at each height were set as objective variables, while air temperature or solar radiation was set as an explanatory variable. Third, another linear regression analysis was conducted with k and air temperature or solar radiation. If an environmental factor exerts a positive effect on vertical variation, R_{wood} in the canopy will show a larger increase than R_{wood} at breast height, which will translate into a positive correlation between k and the environmental factor.

Scaling

At the individual scale ($R_{\text{wood_individual}}$, $\mu\text{mol tree}^{-1} \text{s}^{-1}$), R_{wood} was estimated by multiplying R_{wood} by woody tissue surface area (A_{wood} , m^2) as follows:

$$R_{\text{wood_individual}} = R_{\text{wood}} \times A_{\text{wood}} \quad (3)$$

where A_{wood} was estimated by DBH (cm) using the equation reported by Chambers et al. (2004) obtained in an Amazonian forest as follows:

$$\log(A_{\text{wood}}) = -0.105 - 0.686 \log(\text{DBH}) + 2.208 \log(\text{DBH})^2 - 0.627 \log(\text{DBH})^3 \quad (4)$$

Woody tissue surface area was divided into stem surface area (A_{stem} , m^2) and branch surface area (A_{branch} , m^2) as follows:

$$A_{\text{wood}} = A_{\text{stem}} + A_{\text{branch}} \quad (5)$$

A_{stem} was estimated using the concept of a truncated cone (Yoneda 1993) as follows:

$$A_{\text{stem}} = \pi \times r \times \sqrt{r^2 + h^2} \quad (6)$$

where r and h are the radii at breast height and tree height (m), respectively. A_{branch} was calculated as the difference between A_{wood} and A_{stem} . To examine the potential error in vertical and inter-individual variation on the estimate of $R_{\text{wood_individual}}$, $R_{\text{wood_individual}}$ was estimated by four methods. Method 1 considered both, vertical and inter-individual variation in R_{wood} as follows:

$$R_{\text{wood_individual}} = R_{\text{wood_bh}} \times A_{\text{stem}} + R_{\text{wood_canopy}} \times A_{\text{branch}} \quad (7)$$

where $R_{\text{wood_bh}}$ ($\mu\text{mol m}^{-2} \text{s}^{-1}$) is R_{wood} at breast height and $R_{\text{wood_canopy}}$ ($\mu\text{mol m}^{-2} \text{s}^{-1}$) is R_{wood} in the canopy. Method 2 ignored vertical variation, but considered inter-individual variation as follows:

$$R_{\text{wood_individual}} = R_{\text{wood_bh}} \times A_{\text{wood}} \quad (8)$$

It must be noted that $R_{\text{wood_bh}}$ was obtained in individual trees. Method 3 ignored inter-individual variation, but considered vertical variation as follows:

$$R_{\text{wood_individual}} = R_{\text{wood_average}} \times A_{\text{stem}} + R_{\text{wood_average}} \times k \times A_{\text{branch}} \quad (9)$$

where $R_{\text{wood_average}}$ ($\mu\text{mol m}^{-2} \text{s}^{-1}$) was the average $R_{\text{wood_bh}}$ for the 18 trees studied. Here, k was obtained in individual tree from the measured data at breast height and in the canopy, while $R_{\text{wood_average}}$ was the same value among individuals. Thus, the value of $R_{\text{wood_bh_average}} \times k$ was assumed as $R_{\text{wood_canopy}}$ for each tree. Method 4 considered neither the vertical variation nor the inter-individual variation and was calculated as follows:

$$R_{\text{wood_individual}} = R_{\text{wood_average}} \times A_{\text{wood}} \quad (10)$$

Here, only A_{wood} was obtained in individual tree based on DBH. To compare the effects of vertical and inter-individual variation on estimates of $R_{\text{wood_individual}}$, averages of $R_{\text{wood_individual}}$ estimated using Methods 1, 2, 3 and 4 for the 18 trees were compared. In addition, $R_{\text{wood_individual}}$ using Method 2, 3 and 4 was compared with Method 1.

Results

Vertical variation in wood CO₂ efflux

At breast height, R_{wood} for each tree ranged from 0.63 to 3.42 $\mu\text{mol m}^{-2} \text{s}^{-1}$, with a mean ($\pm\text{SD}$) of $1.74 \pm 0.82 \mu\text{mol m}^{-2} \text{s}^{-1}$; similarly, R_{wood} below the canopy ranged from 0.43 to 3.52 $\mu\text{mol m}^{-2} \text{s}^{-1}$, with a mean of $1.42 \pm 0.78 \mu\text{mol m}^{-2} \text{s}^{-1}$, while R_{wood} in the canopy ranged from 0.30 to 2.62 $\mu\text{mol m}^{-2} \text{s}^{-1}$, with a mean of $1.34 \pm 0.57 \mu\text{mol m}^{-2} \text{s}^{-1}$. The corresponding coefficients of variances at breast height, below the canopy and in the canopy were 48.5%, 54.8% and 42.8%, respectively.

Vertical variation in R_{wood} among trees followed no particular trend (Figure 1). R_{wood} increased significantly with measurement height in only one tree (No. 14, $P < 0.05$, Figure 1), while it decreased in two other trees (Nos. 8 and 11, $P < 0.05$, Figure 1) and was not significantly related with measurement height in the rest of the sampled trees. R_{wood} was not significantly correlated to diameter, $\ln(\text{diameter})$ or measurement height in any of the trees ($P = 0.13$, $P = 0.09$ and $P = 0.89$, respectively, Figure 2). The k values ranged from 0.22 to 2.31, with a mean of 1.01 ± 0.52 and did not relate to either crown ratio or DBH ($P = 0.33$ and $P = 0.95$; see Figure S1 available as Supplementary Data at *Tree Physiology* Online).

The pattern of vertical variation in R_{wood} did not considerably differ with time for the b1 tree (Figure 3). Height was positively related to R_{wood} , and solar radiation and air temperature were negatively related to R_{wood} (ANOVA, $P < 0.01$). No interaction effect of height with either solar radiation or air temperature was observed ($P = 0.15$, $P = 0.39$, respectively). R_{wood} at 40 m height was negatively correlated with air temperature ($P < 0.05$), and R_{wood} at 35 m height was negatively correlated with solar radiation ($P < 0.05$), while R_{wood} at other heights were not related to environmental factors (see Table S1 available as Supplementary Data at *Tree Physiology* Online); further, neither was k related to solar radiation or air temperature ($P = 0.89$, $P = 0.10$, respectively; see Figure S2 available as Supplementary Data at *Tree Physiology* Online).

Scaling whole-tree stem and branch CO₂ efflux

$R_{\text{wood_individual}}$ estimated by Method 1 compared closely to estimates by Methods 2, 3 and 4 ($y = 1.001 \times x$, $R^2 = 0.75$; $y = 1.024 \times x$, $R^2 = 0.55$; and $y = 0.947 \times x$, $R^2 = 0.61$, respectively, Figure 4). The slope of the regression between

$R_{\text{wood_individual}}$ and its estimate by Method 2 was the closest to one, while the coefficient of determination of Method 2 was the highest. Mean R_{wood} as per Methods 1, 2, 3 and 4 were 161.0 ± 127.8 , 170.2 ± 132.8 , 181.3 ± 142.1 and $172.9 \pm 111.8 \mu\text{mol s}^{-1}$ per tree, respectively (Table 2). The ratio of R_{wood} estimates by Methods 2, 3 and 4 to that by Method 1, were 1.06, 1.13 and 1.07, respectively.

Discussion

Vertical variation in R_{wood}

We did not find any clear relationship between R_{wood} and height in the Bornean tropical forest (Figure 1). Many previous studies based on one species only found that R_{wood} increased with stem height, such as in beech (Damesin et al. 2002), in *Chamaecyparis obtusa* (Siebold & Zucc.) Endl. (Araki et al. 2010), in Norway spruce (Tarvainen et al. 2014), in *Styphnolobium japonicum* (L.) Schott (Han et al. 2017) and in Spanish black pine (Martínez-García et al. 2017). Previous studies reported that the vertical variation in R_{wood} was caused by vertical variation in stem growth (Araki et al. 2010, 2015, Tarvainen et al. 2014), respiratory potential (Pruyn et al. 2002, Bowman et al. 2005) and diffused CO₂ transported upwards by flowing sap through the different bark thickness to the atmosphere in the canopy (e.g., Negisi 1979, Bowman et al. 2005, Teskey and McGuire 2007). These factors might be different among tree species and individual size. For example, bark thickness varied greatly within and among species, in addition to size, in a tropical rainforest (Paine et al. 2010). Vertical variation patterns in stem growth among individuals were different because of different light availability for individual trees in a *Chamaecyparis obtusa* stand (Sumida et al. 2013), suggesting that vertical variation in stem growth differs considerably in a tropical rainforest, because multiple canopy layers cause large spatial variation in light conditions. Our sample trees included various species and sizes of trees, which might be a good reason for the absence of a clear trend of vertical variation of R_{wood} .

The present study did not find any clear relationship between k and DBH or crown ratio (Figure 3), which are factors related to vertical variation (Araki et al. 2010, Katayama et al. 2014). Previous studies examined only one or two species (*C. obtusa*, Araki et al. 2010, *D. aromatica* and *S. beccariana*, Katayama et al. 2014), and Katayama et al. (2014) measured relatively large trees with DBH >60 cm. The result reported herein might also be partially explained by the large variability in measurements for various species and sizes of trees in the tropical rainforest.

Vertical variation in R_{wood} was measured at various times of day in the present study, which did not considerably affect the patterns of vertical variation in R_{wood} . Recent studies showed that dissolved CO₂ in the sap flow is transported upward in the

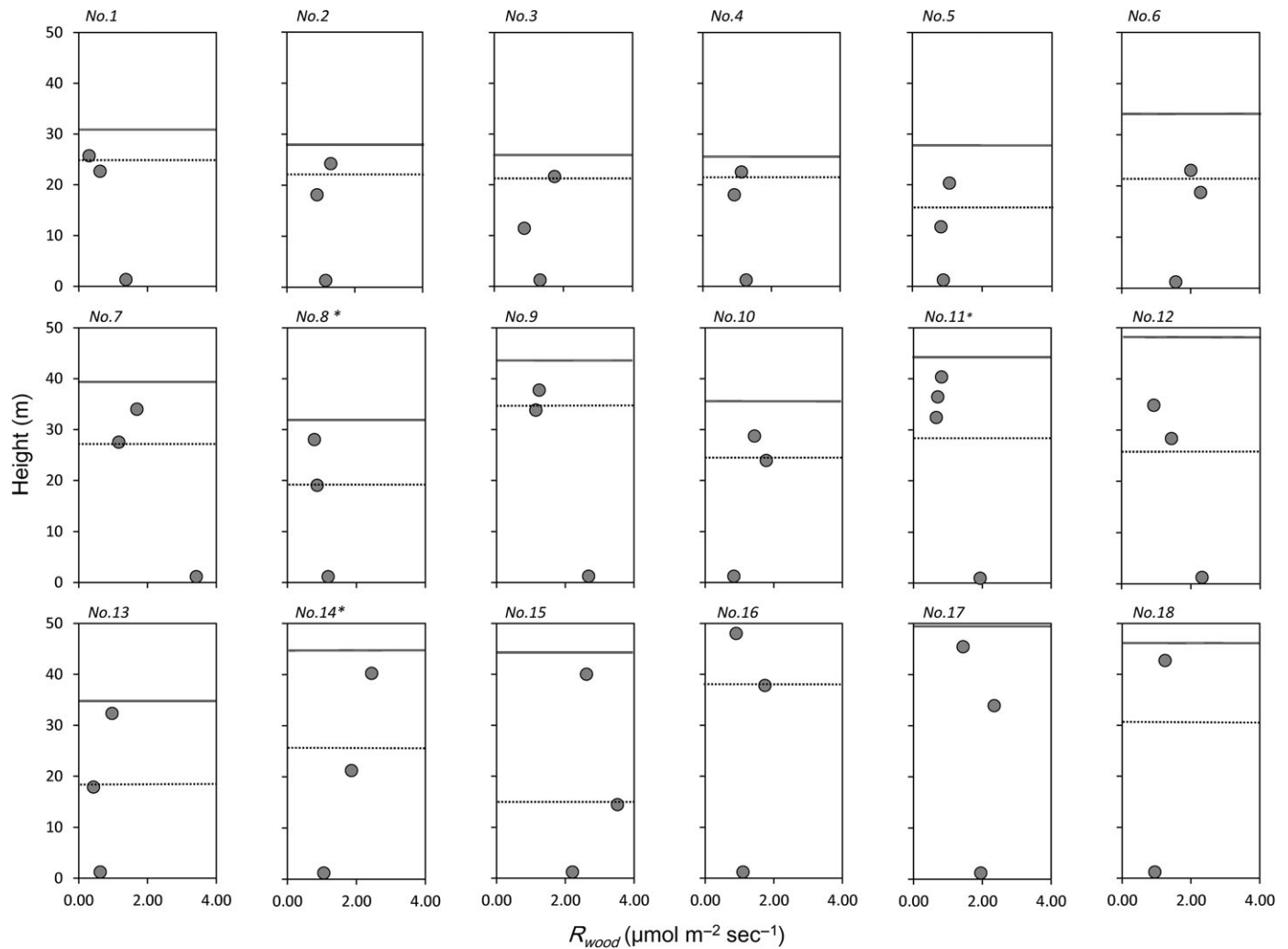


Figure 1. Vertical variation in wood CO₂ efflux (R_{wood}) for each tree. Solid lines and dashed lines show the tree height and the height of the lowest live branches, respectively. Asterisks indicate a significant relationship between R_{wood} and height ($P < 0.05$).

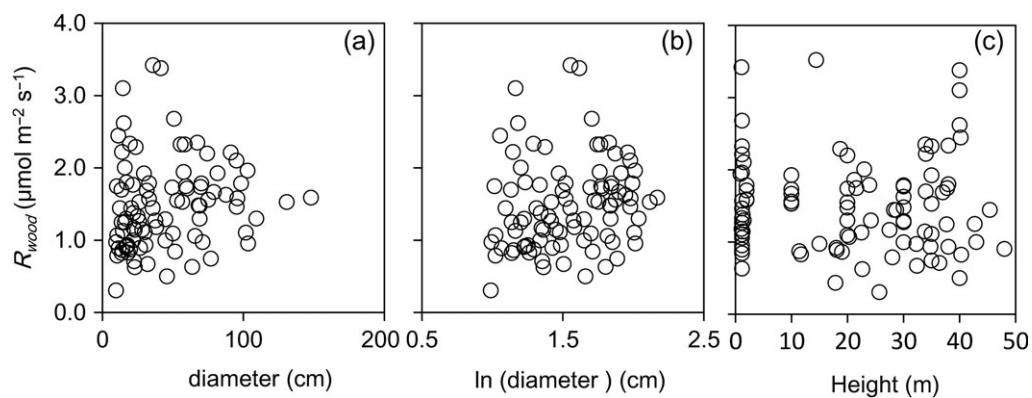


Figure 2. Relationship of R_{wood} with (a) diameter, (b) $\ln(\text{diameter})$ and (c) height.

morning, resulting in lower R_{wood} at breast height and the dissolved CO₂ is diffused in the canopy through thin bark (Teskey and McGuire 2002, Aubrey and Teskey 2009). Generally, sap flow velocity increases with solar radiation and air temperature;

therefore, it was reported that air temperature was negatively correlated with R_{wood} (Kunert 2018). However, the data obtained in the present study did not always show a clear effect of environmental factors on diurnal variation in vertical variation

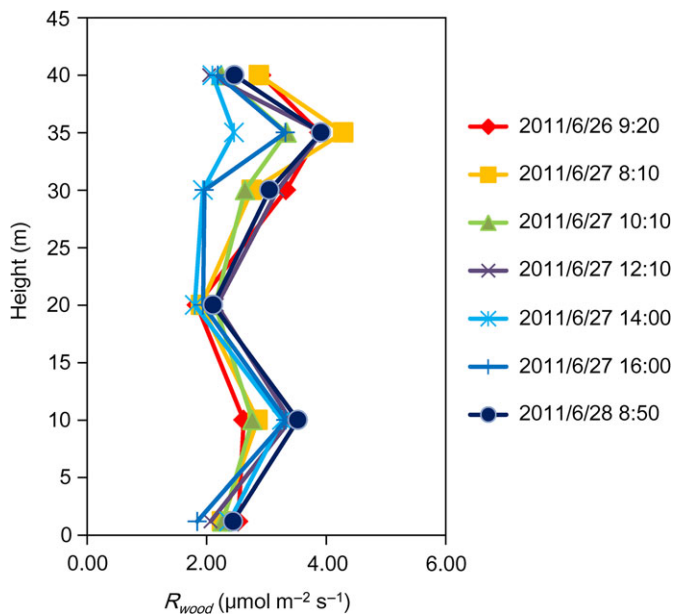


Figure 3. Diurnal variation in vertical variation in R_{wood} for the *S. beccariana* tree. Two-way ANOVA showed that height and measurements time significantly affected R_{wood} , but interaction effect was not significant. Time indicates start of the vertical measurements from the top.

in R_{wood} (Figure 3; see Figure S2 and Table S1 available as Supplementary Data at *Tree Physiology* Online). In addition, measurements of R_{wood} at breast height for five trees showed different patterns of diurnal variation in R_{wood} at breast height in individual trees (Katayama et al. 2016) and R_{wood} was not related to air temperature or solar radiation, except for one tree (see Figure S3 available as Supplementary Data at *Tree Physiology* Online). These results implied that effect of transported CO_2 by sap flow was not clearly observed, which did not significantly affect vertical variation in R_{wood} in the present study.

No measurements of CO_2 efflux from branch might cause no clear vertical variations in the present study. Branch CO_2 efflux was higher at small diameter, resulting in clear vertical variation (Damesin et al. 2002, Cavaleri et al. 2006) because of high nitrogen content, high stem growth rate and thin bark with less resistance to diffusion of CO_2 from tissue. Although studies examining vertical variation in stem CO_2 efflux have increased (Han et al. 2017, Kunert 2018), few studies measured branch CO_2 efflux, probably because of the difficulties of conducting such measurements. Thus, we need to measure vertical variation in R_{wood} including branch.

Effect of vertical variation in $R_{\text{wood_individual}}$

Our study showed that when vertical variation in R_{wood} is ignored, the resulting error in R_{wood} estimation at the individual scale is negligible (Figure 4a and Table 2). Further, the $R_{\text{wood_individual}}$ value estimated by Method 2 was closer to the value estimated by Method 1 than to the value estimated by Method 3, suggesting that if we consider inter-individual

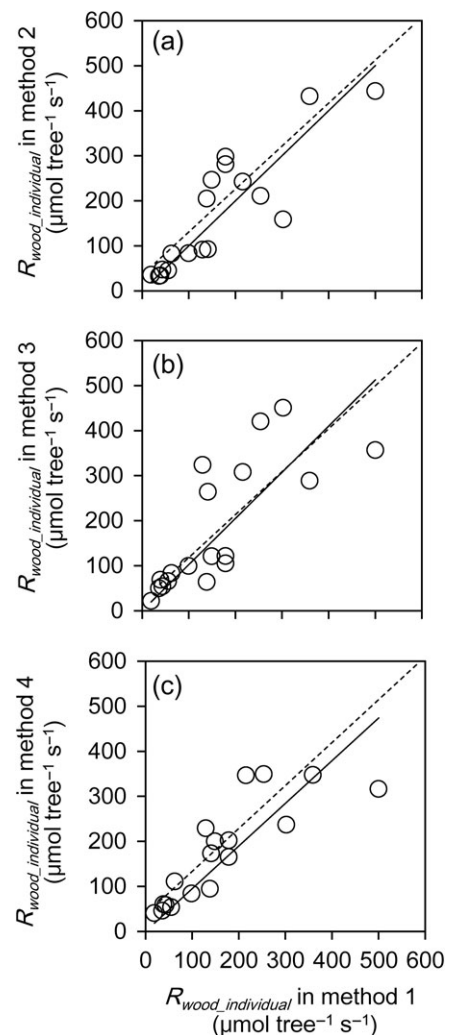


Figure 4. Relationship of $R_{\text{wood_individual}}$ estimated by Method 1, with (a) $R_{\text{wood_individual}}$ estimated in Method 2 ($y = 1.001 \times x$, $R^2 = 0.75$), (b) by Method 3 ($y = 1.024 \times x$, $R^2 = 0.55$) and (c) by Method 4 ($y = 0.947 \times x$, $R^2 = 0.61$). Dotted lines are the 1:1 lines.

variation, but ignore vertical variation, there is less error in the R_{wood} estimate at the individual scale than if we consider vertical variation but ignore inter-individual variation. The coefficient of determination in the relationship with Method 2 was higher than that with Method 3, suggesting that the model considering only inter-individual variation is better than that considering only vertical variation. These results suggest that not making many vertical measurements will not cause large R_{wood} estimate errors, and that individual measurements at breast height are more convenient and thus, preferable for estimating $R_{\text{wood_individual}}$.

Most previous studies reported that R_{wood} increased with stem height and that single breast height measurement of R_{wood} caused a large error in the estimation of R_{wood} at the individual scale (Araki et al. 2010, Tarvainen et al. 2014, Martínez-García et al. 2017). However, those studies examined only one coniferous species and with similar tree size in a temperate area; thus,

Table 2. Mean ($\pm$ SD) $R_{\text{wood_individual}}$ estimates for 18 trees calculated by four different methods and ratios of each mean $R_{\text{wood_individual}}$ estimate to that obtained by Method 1.

Method	Mean $\pm$ SD $R_{\text{wood_individual}}$ ($\mu\text{mol s}^{-1}$ per tree)	Ratio to Method 1
1	161.0 $\pm$ 127.8	
2	170.2 $\pm$ 132.8	1.06
3	181.3 $\pm$ 142.1	1.13
4	172.9 $\pm$ 111.8	1.07

we did not know whether vertical variation in R_{wood} should be considered while estimating R_{wood} at the individual scale, especially in tropical rainforests consisting of a large number of species and sizes of trees. Therefore, we examined the extent of vertical variation in R_{wood} among different species and tree sizes in a Bornean tropical rainforest and found no consistent vertical variation in R_{wood} with height. Furthermore, the absence of a clear-cut tendency in vertical variation in R_{wood} caused little error in the estimates of $R_{\text{wood_individual}}$. Therefore, we speculate that R_{wood} measurements at breast height could hardly cause any large error in R_{wood} estimates at the individual and ecosystem scales in a forest consisting of various tree species of different sizes.

Several uncertainties remain regarding the estimations, such as temporal variation in vertical variation of R_{wood} . Diurnal pattern in R_{wood} varied with stem height and vertical variation pattern differed with time in an Amazonian tree (Kunert 2018), which suggests that diurnal variation in vertical variation is highly complex. This might cause the absence of any consistent patterns of vertical variation in R_{wood} in the present study, although a clear environmental effect on R_{wood} was not observed. Another source of estimate error is CO₂ efflux from branches. Higher variation in branch CO₂ efflux related to diameter was reported (Damesin et al. 2002). The variation will influence $R_{\text{wood_individual}}$ estimate more than stem CO₂ efflux, because average A_{branch} for the 18 trees ($74.4 \pm 46.7 \text{ m}^2 \text{ tree}^{-1}$) was over two-fold greater than A_{stem} ($35.4 \pm 25.1 \text{ m}^2 \text{ tree}^{-1}$). In addition, it is very difficult to estimate surface area of branches at each diameter class. Therefore, it is necessary to examine this variation for accurate wood CO₂ efflux estimation at individual tree scale. Thus, further studies are needed in which continuous vertical measurements of R_{wood} in many diverse tree species of different size, including branches, are conducted, and a systematic effort to understand the mechanism of vertical variation for estimating R_{wood} in a mature forest is made.

Supplementary Data

Supplementary data for this article are available at *Tree Physiology Online*.

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Conflict of interest

None declared.

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